Sarvagya Gupta

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Portfolio: codolio.com/profile/_SarvagyaGupta

Professional Summary

AI/ML Engineering student specializing in building production-ready generative AI and computer vision systems. Proven ability to bridge the gap between experimental models and scalable microservices, with deep expertise in multi-agent LLM architectures, hybrid RAG pipelines, and automated deployments. Passionate about optimizing inference latency and delivering measurable business impact through data-driven engineering.

Technical Skills

Languages: Python, C++, SQL
AI/ML: TensorFlow, PyTorch, Scikit-learn, OpenCV, NLP, Computer Vision, Model Evaluation
GenAI & LLMs: LangChain, LangGraph, Ollama, RAG, Vector Databases (Qdrant, Pinecone), RAGAS
Backend & MLOps: FastAPI, Docker, AWS, Linux, CI/CD, GitHub Actions, MLflow, REST APIs
Dev Tools: Git, Pytest, Weights & Biases, Matplotlib, Seaborn

Professional Experience

Machine Learning Intern | SkillCraft Technology *June 2025 – August 2025*

- Engineered a core TensorFlow image classification API to automate production workflows, scaling it to handle **10,000+** daily inference requests by containerizing the model behind FastAPI.
- Automated model deployment cycles via a GitHub Actions CI/CD pipeline, utilizing Docker and TensorFlow Serving to reduce version rollback times by **40%** (from 30m to 18m).
- Fortified pipeline reliability by integrating Pytest and Bandit static analysis, catching **15+** critical logic and security vulnerabilities.

Projects

Multi-Agent Research System | GitHub | Live Demo *May 2026 – June 2026*
Python, LangGraph, FastAPI, AWS EC2, LLaMA-3 70B

- Architected a 4-agent LLM research pipeline (LangGraph, LLaMA-3) that automates web search and synthesis, reducing average research time from 30 minutes to 80 seconds (**22x speedup**).
- Improved report accuracy to a **66.5%** fact-check pass rate by implementing a dedicated verification agent to catch hallucinations before final synthesis.
- Engineered a production-ready FastAPI backend deployed on AWS EC2, integrating Mem0 for persistent session memory to support contextual follow-up queries.

RAG Eval Studio | GitHub | Live Demo *March 2026 – April 2026*
Python, Qdrant, RAGAS, LangSmith, LangChain, AWS S3

- Engineered a hybrid RAG system (Qdrant, BM25) to query **5,500+** pages of SEC filings, achieving **0.64** context precision and **0.79** answer relevancy using a 512-token chunking strategy.
- Eliminated self-evaluation bias by benchmarking the generator against a 70B judge model using RAGAS, tracing all queries in LangSmith for performance observability.
- Automated document ingestion from AWS S3 and established a CI/CD pipeline with GitHub Actions to run unit tests on every push.

Plant Disease Classification | GitHub | Live Demo *September 2025 – December 2025*
Python, CNNs, TensorFlow, FastAPI, Docker

- Fine-tuned a MobileNetV2 architecture on **3,000** PlantVillage images using transfer learning, achieving **96.4%** test accuracy and a **0.964** F1-score across 3 disease classes.
- Optimized model generalizability by engineering a data augmentation pipeline (rotation, zoom, contrast) and deployed the **2.6M parameter** model via TensorFlow Serving.
- Developed an async FastAPI microservice that wraps CNN predictions and feeds them to an LLM chatbot, delivering structured agricultural treatment plans in ∼**160ms**.

Certifications & Achievements

- Ranked **5th** at CodeSrijn 2026 National AI Hackathon among 50+ teams, built Datafort AI. Link
- Solved **700+ DSA problems** across LeetCode, Codechef, and competitive programming platforms. Profile
- CS50 AI with Python – Harvard University (2025). Certificate
- Introduction to Generative AI – Google (2023). Certificate
- Introduction to LLMs – Google (2023). Certificate

Education

B.Tech in Computer Science and Engineering *July 2023 – May 2027*
Jaypee University of Engineering and Technology, Guna CGPA: 8.2 / 10
Relevant Coursework: Machine Learning, Artificial Intelligence, Linear Algebra, Probability & Statistics, Data Structures & Algorithms